i. Greater adaptability and successful colonisation on every type of habitat.

ii. The relative ease of cultivation

iii. Tillering property that produce more branches which results in higher yield per unit area.

iv. Compact and dry grains that they can be easily handled, transported and stored without undergoing spoilage.

v. High caloric value that provides energy.

The nutrients provided by cereals include carbohydrates, proteins, fibres and a wide range of vitamins and minerals. Cereals can be classified into two different types based on their size namely Major Cereals and Minor Cereals.

Major Cereals

Rice / Paddy

Botanical name : *Oryza sativa*

Paddy is a semi-aquatic crop and is grown in standing water. It is an important food crop of the world, occupying the second position in terms of area under cultivation and production, next to wheat. Rice is the chief source of carbohydrate.

Origin and Area of cultivation

South East Asia is considered as the center of origin of rice. Earliest evidences of rice cultivation have been found in China, India and Thailand. It is mainly cultivated in Delta and irrigated regions of Tamil Nadu.

Uses

Rice is the easily digestible calorie rich cereal food which is used as a staple food in Southern and North East India. Various rice products such as **Flaked rice** (Aval), **Puffed rice / parched rice** (Pori) are used as breakfast cereal or as snack food in different parts of India.

Rice bran oil obtained from the rice bran is used in culinary and industrial purposes.

Husks are used as fuel, and in the manufacture of packing material and fertilizer.

International Rice Research Institute (IRRI)



International Rice Research Institute (IRRI) is located in Los Banos, Manila the capital city of Philippines. This is the only institute in the world which exclusively carries out research as on rice. IRRI aims to improve livelihoods and nutrition, abolishing poverty, hunger, and malnutrition. Whatever IR rice varieties available in the world are developed through rice breeding programme and released by IRRI. Till date IRRI has produced 843 rice varieties that have been released in 77 countries. IR8 is a high-yielding semi-dwarf rice variety developed by IRRI in the early 1960s and it is called as miracle rice, much celebrated for fighting famine. Another variety to mention is IR36 which is a semi-dwarf variety that proved highly resistant to a number of insect pests and diseases that raised farmers' rice yields and brought down the prices of the staple food in Asian families. The International Rice Gene bank of IRRI has a collection of more than 117 000 types of rice, comprising of modern and traditional varieties including wild relatives of Paddy.


Rice


Wheat


Maize

Figure 10.1: Major Cereals

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